The Subgraph Query Problem for K_6

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Abstract

Let the edges of an infinite Erdős–Rényi random graph be hidden. In each turn Builder chooses a pair of vertices and learns whether the pair is an edge. Given a fixed graph H, the Subgraph Query Problem asks how many turns Builder needs in order to find a copy of H with probability at least one half. Conlon, Fox, Grinshpun, and He determined this number for cliques of order at most five and conjectured its order for every clique. In particular, their conjecture predicts that

$$f(K_6, p) = \Theta(p^{-10/3}).$$

We prove this prediction. The lower bound is obtained by ordering the fifteen edges in every completed copy of K_6 and studying the graph formed by the first nine successful edges. The recursive estimates of Conlon, Fox, Grinshpun, and He deal with all but three of the possible nine-edge graphs. Two of the exceptions are controlled by exposing the random host in advance. The third is the four-page book $K_2 \vee \overline{K_4}$; although this graph may occur too often, the next successful edge in a completed K_6 gives a ten-edge graph that can again be counted. Together with the branch-and-fill strategy for the upper bound, this yields the claimed order of magnitude.

1 Introduction

Let $p \in (0,1)$, and suppose that Builder plays the following one-player game on an initially hidden random graph $G(\mathbb{N}, p)$. In each turn he queries one pair of vertices and is told whether that pair is an edge. We say that an edge is *successfully built* when the answer is positive, and, following [3], use the words "query" and "build" interchangeably when no confusion can arise. Builder's aim is to build a copy of a prescribed graph H in as few turns as possible.

For a fixed graph H, let $f(H,p)$ be the minimum number N for which Builder has a strategy that builds a copy of H in at most N turns with probability at least $1/2$. There is no advantage in allowing Builder private randomness: after fixing his random seed, some deterministic strategy has success probability at least the average. We may also assume that no pair is queried twice, since a repeated query reveals nothing new.

The Subgraph Query Problem was introduced by Conlon, Fox, Grinshpun, and He in connection with online Ramsey numbers [3]. They conjectured that, for every $m \geq 4$,

$$f(K_m, p) = 2^{o(m)} p^{-2m/3 + c_m}, \qquad c_m = \begin{cases} \dfrac{m}{2m-3}, & m \equiv 0 \pmod 3, \\ \dfrac{2}{3}, & m \equiv 1 \pmod 3, \\ \dfrac{2m+8}{6m-3}, & m \equiv 2 \pmod 3. \end{cases} \tag{1.1}$$

For $m = 6$, the exponent in (1.1) is 10/3. The results of [3] give

$$f(K_3, p) = \Theta(p^{-3/2}), \qquad f(K_4, p) = \Theta(p^{-2}), \qquad f(K_5, p) = \Theta(p^{-8/3}),$$

as well as

$$f(K_6, p) = \Omega(p^{-13/4}) \qquad \text{and} \qquad f(K_6, p) = O(p^{-10/3}).$$

The upper bound here is the $m = 6$ instance of their branch-and-fill construction. Thus the remaining problem is to improve the lower exponent from 13/4 to 10/3.

Theorem 1.1. *There are absolute constants $0 < c < C < \infty$ such that, for all sufficiently small $p > 0$,*

$$cp^{-10/3} \le f(K_6, p) \le Cp^{-10/3}.$$

Equivalently, $f(K_6, p) = \Theta(p^{-10/3})$ as $p \to 0$.

The result has also been formalized in Lean 4 [15]; Appendix B records the precise statement and reproduction details.

The proof

We first describe the main ideas. As in [3], let $t(H, p, N)$ be the largest expected number of copies of H that Builder can construct in N turns. The function t satisfies two useful recursive inequalities. One distinguishes the last edge built in a copy of H; the other first chooses the image of an adjacent pair of vertices and then counts the remaining graph. Iterating these inequalities gives lower bounds for $f(H, p)$.

For the present problem, put

$$N = \lfloor \lambda p^{-10/3} \rfloor, \qquad 0 < \lambda \le 1.$$

An N-turn strategy successfully builds about pN edges. If the whole random graph on the at most $2N$ vertices mentioned by Builder is exposed in advance, a pair and a triple of vertices have about p^2N and p^3N common neighbors, respectively. At the above value of N, these three quantities have orders

$$pN = \lambda p^{-7/3}, \qquad p^2N = \lambda p^{-4/3}, \qquad p^3N = \lambda p^{-1/3}.$$

They all tend to infinity, so the estimates we need hold simultaneously for every graph that Builder might construct.

Now fix the order in which the fifteen edges of a particular labeled K_6 are queried. If J_r is formed by the first r edges in this order, then after J_r has been built the remaining $15 - r$ edges must all be successful. This costs a factor p^{15-r}, even though Builder's later choices may depend on the answers he receives. For $r = 9$, the two recursive inequalities for t give the required bound except for three graphs:

$$H_3, \qquad (K_5 - e) \sqcup K_1, \qquad B_4 := K_2 \vee \overline{K_4}.$$

The first two can be counted inside the random host. The last graph is the four-page book. There may be too many copies of this book for the same argument to work directly. However, its six missing edges are precisely the pairs between the four pages. The tenth edge of any completed K_6 therefore creates

$$Q := K_2 \vee (K_2 \sqcup 2K_1);$$

see Figure 1. and this ten-edge graph has few enough copies. Summing over all edge orders shows that the probability of success in N turns is at most $C_2\lambda^2 + C_3\lambda^3 + o(1)$. Choosing λ sufficiently small proves the lower bound.

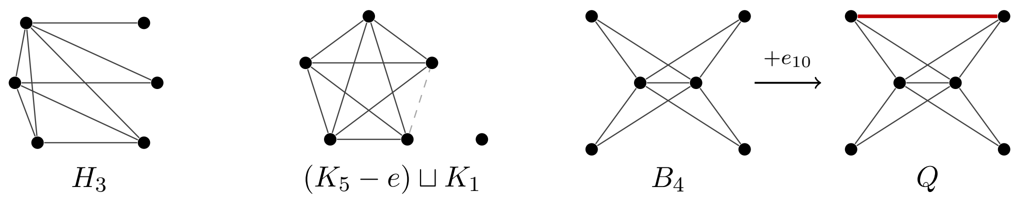

Figure 1: The three exceptional nine-edge prefixes and the ten-edge graph Q. The dashed pair is absent from the nine-edge prefix (it is not meant to record a negative query answer). The highlighted edge is one representative choice for the tenth edge after B_4; all six pairs among the four pages give isomorphic copies of Q.

Related work

A related query problem was studied by Ferber, Krivelevich, Sudakov, and Vieira, who considered finding Hamilton cycles and long paths in a finite random graph using few adjacency queries [8, 9]. Here the target graph is fixed and the ambient random graph is much larger. The formulation used in this paper is due to Conlon, Fox, Grinshpun, and He, who showed that it is closely connected with the online Ramsey game against a random Painter [3]. For further background on online Ramsey games, see Beck [2], Conlon [4], Kurek and Ruciński [16], and Friedgut, Kohayakawa, Rödl, Ruciński, and Tetali [10]. More recently, Guo and Warnke obtained alteration bounds for related games [11].

Alweiss, Ben Hamida, He, and Moreira proved general upper bounds for fixed targets, including a strict improvement over the elementary degeneracy bound, and exhibited fixed graphs whose query complexity is not a constant power of p^{-1} [1]. Their results show that the clean power laws for small cliques are not representative of every target graph. They do not determine the clique exponent in the first unresolved case $m = 6$.

There is also a finite-n literature in which the target clique grows with the ambient graph and the algorithm is allowed a bounded number of query rounds. Feige, Gamarnik, Neeman, Rácz, and Tetali initiated this direction for large cliques in $G(n, 1/2)$ [7]; Csóka and Pongrácz sharpened several bounds and considered dense subgraphs [5]. Query limits for detecting a planted, rather than naturally occurring, subgraph form another distinct model [18, 13, 12]. These works share the theme of restricted access to a random graph, but their asymptotic regimes and success events differ from the fixed-H, $p \to 0$ problem here. To the best of our knowledge, none of these developments determined $f(K_m, p)$ for any new fixed clique order $m \geq 6$; thus K_6 remained the first unresolved case of (1.1).

Organization

The rest of the paper is organized as follows. In Section 2, we reduce the game to a finite vertex set and prove the two recursive bounds for $t(H, p, N)$. The random-host estimates are proved in Section 3, and the finite calculation for nine-edge graphs is given in Section 4. Sections 5 and 6 contain the lower and upper bounds, respectively. The finite calculation and the formalization are discussed further in the appendices.

Unless otherwise indicated, all asymptotics are as $p \to 0$. We do not attempt to optimize the absolute constants in the proof.

2 Recursive graph building

Following [3], our lower bound will be expressed in terms of the expected number of copies that Builder can construct. We begin with a harmless reduction. Although the game is played on a countably infinite vertex set, a strategy lasting N turns can mention at most $2N$ vertices. We need this observation in a slightly precise form because some of the prefix graphs considered later have isolated vertices.

2.1 A finite vertex set

Write $[2N] = \{0, 1, \ldots, 2N - 1\}$. A strategy on $[2N]$ chooses an unqueried pair in each turn, using the previous answers to choose the next pair. Equivalently, one may assign independent Bernoulli(p) variables to all pairs of $[2N]$ at the outset and reveal a variable when Builder queries its pair. Vertices that Builder never mentions are still present in the ambient set and will count as isolated vertices when we embed a prefix graph.

Proposition 2.1 (finite-board reduction for K_6)**.** *For every $N \geq 0$ and $p \in [0, 1]$, the supremum of the success probabilities in the Subgraph Query Game for K_6, over strategies using at most N turns, is unchanged if Builder is required to use the vertex set $[2N]$. The same is true if randomized strategies are allowed.*

Proof. We have already observed that repeated pairs and loops may be discarded. Also, a strategy that stops early may be padded with arbitrary unused pairs; this does not destroy a clique that has already been built. For $N \geq 1$, there are enough pairs in $[2N]$ to do this, while $N = 0$ is immediate.

Now fix a deterministic strategy on the countably infinite vertex set. Name the vertices in the order in which Builder first mentions them. Whenever a query introduces one or two new vertices, give them the next unused names in $[2N]$, breaking a tie by any fixed ordering of the original vertex set. After s turns, no more than $2s$ vertices have been named.

It remains only to note that this relabeling defines one strategy, rather than a different strategy for every possible play. Indeed, two plays with the same first s answers lead the original strategy to the same first s queries. They therefore give the same relabeling of all vertices seen so far, and hence the same relabeled next query. Thus the construction is consistent on the whole decision tree. For every complete sequence of answers, the graph successfully built by the new strategy is isomorphic to the graph built by the old one on the vertices that were mentioned. The answers also have the same independent Bernoulli law. The two strategies therefore have the same success probability.

Conversely, any strategy on $[2N]$ can plainly be used on the infinite vertex set. This proves the deterministic statement. A randomized strategy is a mixture of deterministic ones, and one member of the mixture has success probability at least its average. $\square$

For the rest of the lower-bound proof, all graphs live on $[2N]$. In particular, if a prefix is $(K_5 - e) \sqcup K_1$, the isolated vertex may be sent to any of the $2N$ ambient vertices, whether or not Builder has mentioned that vertex.

For a simple labeled graph H, let $N_H(F)$ be the number of injective, not necessarily induced, embeddings of H into F. Define

$$t(H, p, N) = \max_{\mathcal{A}} \mathbb{E}_p \, N_H(F_N(\mathcal{A})), \tag{3.1}$$

where the maximum is over deterministic N-turn strategies on $[2N]$, and $F_N(\mathcal{A})$ is the graph of the edges successfully built by $\mathcal{A}$. The maximum exists because the decision tree is finite. We count labeled embeddings throughout. This changes only constant factors, but it makes the recursive inequalities below exact and keeps isolated vertices on the same footing as the others.

2.2 Two recursive bounds

For an edge e of H, write $H-e$ for the graph obtained by deleting that edge. If $u, v \in V(H)$, write $H - \{u, v\}$ for the graph obtained by deleting the two vertices. Let $Y = e(F_N)$. Since Builder queries a new pair in every turn, each answer is positive with probability p, independently of the previous answers. Hence

$$Y \sim \mathrm{Bin}(N, p),$$

even though Builder chooses the pairs adaptively. For a fixed $\kappa > 1$, put

$$R_\kappa = \lfloor \kappa p N \rfloor + 1, \qquad \tau_\kappa(p, N) = (2N)^6 \mathbb{P}(\mathrm{Bin}(N, p) \geq R_\kappa). \tag{3.2}$$

For example, with $\kappa = 8$,

$$\mathbb{P}(\mathrm{Bin}(N, p) \geq R_8) \leq \binom{N}{R_8} p^{R_8} \leq (e/8)^{R_8} \leq 2^{-R_8}, \tag{3.3}$$

up to the harmless effect of the integer cutoff. In particular, τ_8 beats every fixed power of p at the scale $N = \lfloor \lambda p^{-10/3} \rfloor$, because there $pN = \Theta(p^{-7/3}) \gg \log(1/p)$.

Lemma 2.2 (expected-copy recurrences)**.** *For every labeled graph H on at most six vertices,*

$$t(H, p, N) \leq p \sum_{e \in E(H)} t(H - e, p, N), \tag{3.4}$$

$$t(H, p, N) \leq 2\kappa p N \, t(H - \{u, v\}, p, N) + \tau_\kappa(p, N) \tag{3.5}$$

for every $uv \in E(H)$. In (3.5), $H - \{u, v\}$ is the induced graph remaining after the two pattern vertices are removed.

Proof. Fix any strategy for Builder. For each copy of H that is built, distinguish the edge of H whose image was queried last. If this distinguished edge is e, then before its final query Builder has already built a copy of $H - e$, and the last answer is positive with probability p. Summing over the possible choices of e and then maximizing over Builder's strategy gives (3.4).

For the second inequality, an embedding of $H - \{u, v\}$, together with the successfully built image of uv and one of its two orientations, determines at most one embedding of H. Consequently, for every outcome of the game,

$$N_H(F_N) \leq 2Y \, N_{H - \{u,v\}}(F_N).$$

On the event $Y \leq \kappa p N$, we take expectations and use the definition of t. On the complementary event we use the crude estimate $N_H(F_N) \leq (2N)^6$. This proves (3.5). $\square$

For later quantitative bookkeeping, the concrete choice $\kappa = 8$ gives

$$t(H, p, N) \leq 16pN \, t(H - \{u, v\}, p, N) + (2N)^6 \, 2^{-R_8}. \tag{3.6}$$

The factor 2 in (3.5) is simply the number of orientations of the image edge. In [3, Lemma 25], the same recursion is written with a different copy-counting convention and an asymptotic factor $1 + o(1)$. We keep the binomial tail explicit because we will apply the inequality several times. Since only finitely many applications are needed, every resulting tail is still $o(1)$ at the scale considered below, even after multiplication by a fixed power of N.

2.3 Stopping at an ordered prefix

The function $t(H, p, N)$ does not remember the order in which the edges of H were queried. The following observation allows us to recover just the amount of order information that is needed. It is here that adaptivity must be treated with some care.

Let $\pi = (e_1, \dots, e_{15})$ be an ordering of the edges of a labeled K_6, and let $J_r(\pi)$ be the spanning six-vertex graph with edge set $\{e_1, \dots, e_r\}$. For a strategy, let X_π count completed labeled copies of K_6 whose edges were queried in relative order π.

Lemma 2.3 (ordered-prefix bound). *For $1 \le r \le 15$,*

$$\mathbb{E}X_\pi \le p^{15-r} t(J_r(\pi), p, N). \tag{3.7}$$

The statement remains valid when $J_r(\pi)$ has isolated vertices.

Proof. Fix an injection $\phi : V(K_6) \to [2N]$. Stop when the pair $\phi(e_r)$ is queried, provided that it is successful, that the images of $e_1, \dots, e_{r-1}$ have already been built, and that none of $\phi(e_{r+1}), \dots, \phi(e_{15})$ has yet been queried. We do not insist that the first $r-1$ edges appeared in their prescribed order. Dropping this requirement can only increase the number of possible prefixes.

At the stopping time, all $15-r$ remaining pairs are still unqueried. Builder may use future answers when deciding whether and when to query them, but whenever one of these pairs is first queried its conditional probability of success is p. Iterating this observation at the successive query times shows that the conditional probability of building all the remaining edges in the required order is at most p^{15-r}. If Builder never queries one of them, the desired event simply does not occur.

Every completed copy counted by X_π gives exactly one such stopped prefix. For any fixed play, the number of stopped prefixes is at most the number of embeddings of $J_r(\pi)$ in the final graph F_N. This remains true when $J_r(\pi)$ has an isolated vertex, since its image is chosen from the fixed set $[2N]$. Summing over the possible prefixes and taking expectations proves (3.7). ☐

3 Counting inside the random host

Fix $0 < \lambda \le 1$ and put

$$N = \lfloor \lambda p^{-10/3} \rfloor, \qquad n = 2N.$$

All asymptotics in this section are as $p \to 0$, with λ fixed. Before Builder begins, expose a graph $\Gamma \sim G(n, p)$ on the entire vertex set. When Builder queries a pair, reveal the corresponding edge of Γ. This gives exactly the same distribution as the original game, but it has one important advantage: the graph F_N successfully built by any strategy is a subgraph of the same host Γ. It will therefore be enough to prove estimates that hold simultaneously for all sparse subgraphs of Γ.

Since Builder makes exactly N distinct queries,

$$e(F_N) \sim \mathrm{Bin}(N, p).$$

Thus

$$\mathbb{P}(e(F_N) > M_0) \le \exp(-\Omega(pN)), \qquad M_0 := \lceil 8pN \rceil. \tag{4.1}$$

For a graph G, let $\Delta_j(G)$ denote its maximum j-codegree:

$$\Delta_j(G) := \max_{\substack{S \subseteq V(G) \\ |S|=j}} \left| \bigcap_{v \in S} N_G(v) \right|.$$

Proposition 3.1 (uniform host estimates)**.** *For every fixed $\lambda > 0$ and $K > 0$, with probability $1 - O(p^K)$, the random graph Γ has the following properties simultaneously:*

(i) $\Delta_2(\Gamma) \le Cp^2 N$ *and* $\Delta_3(\Gamma) \le C\lambda^{-1} p^3 N$.

(ii) *Every subgraph $F \subseteq \Gamma$ with $e(F) \le M_0$ satisfies*

$$N_{K_4}(F) \le Ce(F).$$

(iii) *Every such F satisfies*
$$N_{H_3}(F) \le Cp^2 N M_0 e(F) \le Cp^4 N^3.$$

(iv) *Every such F satisfies*

$$N_{(K_5-e)\sqcup K_1}(F) \le C\lambda^{-1} p^3 N^2 e(F) \le C\lambda^{-1} p^4 N^3.$$

(v) *For $Q = K_2 \vee (K_2 \sqcup 2K_1)$, every such F satisfies*

$$N_Q(F) \le Cp^4 N^2 e(F) \le Cp^5 N^3.$$

Apart from the displayed λ^{-1}, the constant C may depend on K, but not on λ; the implicit constant in the failure probability may depend on the fixed λ. The deliberately loose triple-codegree cutoff isolates the only displayed coefficient blow-up as $\lambda \downarrow 0$, which is sufficient because λ is fixed before $p \to 0$.

The powers in this proposition are arranged for the ordered-prefix argument. The bounds for H_3 and $(K_5 - e) \sqcup K_1$ will be multiplied by p^6, while the bound for Q will be multiplied by p^5. After substituting $N = \lambda p^{-10/3}$, all three contributions are small when λ is small.

We prove the proposition in two steps. First we show that every sparse subgraph of Γ has a useful degeneracy ordering. We then use that ordering, together with the pair and triple codegrees of the host, to count the four graphs that occur in the proof.

3.1 Sparse subgraphs of the host

Choose a sufficiently large absolute constant A, and set

$$D = \left\lceil A\sqrt{pM_0} \right\rceil = \Theta(p^{-2/3}).$$

We require two uniform properties of Γ. Since they hold for every subgraph and every small vertex set, they will automatically apply inside all links of Γ.

The first lemma rules out a dense core in any subgraph with at most M_0 edges. The second says that even the small sets arising as forward neighborhoods in a degeneracy ordering span only linearly many host edges.

Lemma 3.2. *With failure probability smaller than every fixed power of p, every subgraph of Γ having at most M_0 edges is D-degenerate.*

Proof. If a subgraph $J \subseteq \Gamma$ with $e(J) \le M_0$ is not D-degenerate, then some vertex set U of size k satisfies

$$D \le k \le \frac{2M_0}{D}, \qquad e_\Gamma(U) \ge \frac{Dk}{2}.$$

For a fixed k, a union bound gives

$$\begin{aligned}\mathbb{P}\left(\exists U:\ |U| = k,\ e_\Gamma(U) \ge Dk/2\right) &\le \binom{n}{k}\binom{\binom{k}{2}}{\lceil Dk/2\rceil} p^{\lceil Dk/2\rceil} \\ &\le \left(\frac{en}{k}\right)^k \left(\frac{ekp}{D}\right)^{Dk/2}. \end{aligned} \tag{4.2}$$

Throughout the permitted range,

$$\frac{kp}{D} \le \frac{2pM_0}{D^2} \le \frac{2}{A^2}.$$

Choose A so large that the second base in (4.2) is less than $1/2$. Since $D \gg \log N$, the second factor dominates the entropy term $(en/k)^k$. Summing over k gives $\exp(-\Omega(D^2))$, which is smaller than every fixed power of p. ☐

Lemma 3.3. *For every fixed $K > 0$, there is a constant $L = L(K)$ such that, with probability $1 - O(p^K)$, every set U with $|U| \le D$ spans at most $L|U|$ edges in Γ.*

Proof. For $k \ge 2L + 1$,

$$\begin{aligned}\mathbb{P}\left(\exists U:\ |U| = k,\ e_\Gamma(U) \ge Lk\right) &\le \binom{n}{k}\binom{\binom{k}{2}}{Lk} p^{Lk} \\ &\le \left[\frac{en}{k}\left(\frac{ekp}{2L}\right)^{L}\right]^k. \end{aligned} \tag{4.3}$$

The bracket is at most a constant multiple of

$$np^L D^{L-1}$$

which at the present scale is

$$O\Big(p^{(L-8)/3}\Big).$$

For $k \le 2L$, the event $e_\Gamma(U) \ge Lk$ is impossible. Taking L sufficiently large and summing (4.3) over $k \le D$ gives failure probability $O(p^K)$. ☐

3.2 Counting K_4, H_3, $K_5 - e$, and Q

We call the graph obtained from a triangle by adding one pendant edge a *paw*, and write $\mathrm{Paw}(J)$ for the number of its labeled copies in a graph J.

Proof of Proposition 3.1. For a fixed pair or triple S, its common-neighbor count is binomial with mean at most $np^{|S|}$. Chernoff's inequality [14], followed by a union bound over at most n^3 sets S proves (i): the relevant means have orders $p^2N = \Theta(p^{-4/3})$ and $p^3N = \Theta(p^{-1/3})$, both much larger than $\log n$. Replacing the triple threshold by the larger $C\lambda^{-1}p^3N$ only strengthens this tail estimate.

Now fix a host satisfying Lemmas 3.2 and 3.3, and let $F \subseteq \Gamma$ have $e(F) \le M_0$. Take a D-degeneracy ordering of F, and let B_x be the set of later F-neighbors of x. Assign each copy of

K_4 to its earliest vertex x. Its other three vertices form a triangle in $\Gamma[B_x]$. Since every subset of B_x spans at most L times its order, $\Gamma[B_x]$ is $2L$-degenerate and hence has $O_L(|B_x|)$ triangles. Therefore

$$N_{K_4}(F) \le C_L \sum_x |B_x| = C_L e(F). \tag{4.4}$$

This proves (ii).

The graph H_3 has vertices $a_1, a_2, a_3, b_1, b_2, b_3$, where $\{a_1, a_2, a_3\}$ is a clique, $\{b_1, b_2, b_3\}$ is independent, and $a_i b_j$ is an edge exactly when $i \le j$. Its vertex a_1 has degree five. After deleting a_1, the vertices a_2, a_3, b_2, b_3 form a paw, while b_1 is isolated. Consequently,

$$N_{H_3}(F) \le C \sum_v d_F(v)\ \mathrm{Paw}\big(F[N_F(v)]\big). \tag{4.5}$$

Put $J_v = F[N_F(v)]$. Applying the same degeneracy charge to this subgraph makes the estimate explicit: order J_v degenerately, assign a triangle to its earliest vertex x, and observe that its other two vertices form an edge in a set of at most D later neighbors. By Lemma 3.3, summing these edge counts gives at most $Le(J_v) \le LM_0$ triangles. Moreover,

$$\Delta(J_v) \le \Delta_2(\Gamma) \le Cp^2 N.$$

Every paw is obtained from a triangle by choosing one of its three vertices and one additional neighbor, so

$$\mathrm{Paw}(J_v) \le 3\Delta(J_v) N_{K_3}(J_v) \le Cp^2 N M_0.$$

Using $\sum_v d_F(v) = 2e(F)$ in (4.5) proves (iii).

For (iv), first omit the isolated vertex. Let r_T be the number of common F-neighbors of a triangle T. In $K_5 - e$, the three degree-four vertices form a triangle and the other two vertices are common neighbors of that triangle. Thus

$$\begin{aligned} N_{K_5-e}(F) &\le C \sum_{T \in K_3(F)} r_T^2 \\ &\le C\Delta_3(\Gamma) \sum_T r_T \\ &\le C\lambda^{-1} p^3 N\, N_{K_4}(F) \le C\lambda^{-1} p^3 N e(F), \end{aligned}$$

because $\sum_T r_T$ is a fixed labeled multiple of the K_4 count. There are at most $2N$ choices for the isolated image vertex, which proves (iv).

Finally, let $C_F(uv) = N_F(u) \cap N_F(v)$. A copy of $Q = K_2 \vee (K_2 \sqcup 2K_1)$, rooted at its central edge uv, is specified by an edge inside $F[C_F(uv)]$ and two further vertices of $C_F(uv)$. Hence

$$\begin{aligned} N_Q(F) &\le C \sum_{uv \in E(F)} e\big(F[C_F(uv)]\big) |C_F(uv)|^2 \\ &\le C\Delta_2(\Gamma)^2 \sum_{uv \in E(F)} e\big(F[C_F(uv)]\big) \\ &= O\Big(p^4 N^2\, N_{K_4}(F)\Big) = O(p^4 N^2 e(F)). \end{aligned}$$

The penultimate estimate uses

$$\sum_{uv \in E(F)} e\big(F[C_F(uv)]\big) = \Theta(N_{K_4}(F)),$$

again with only a fixed labeling factor. This proves (v). $\square$

Corollary 3.4 (expected counts for Builder)**.** *For every N-turn strategy on* $[2N]$,

$$t(H_3, p, N) = O(p^4 N^3) + o(1),$$
$$t((K_5 - e) \sqcup K_1, p, N) = O(\lambda^{-1} p^4 N^3) + o(1),$$
$$t(Q, p, N) = O(p^5 N^3) + o(1).$$

Proof. On the event in Proposition 3.1 and the event $e(F_N) \le M_0$, apply its deterministic bounds to F_N.

On the complementary event, bound every six-vertex count by $(2N)^6$. The host failure probability can be made $O(p^K)$ for arbitrary K, and (4.1) fails with probability $\exp(-\Omega(pN))$. Taking $K > 30$ makes the exceptional contribution $o(1)$ relative to all displayed bounds. □

4 The nine-edge graphs

We now apply the two recursive bounds from Section 2 to the possible nine-edge prefixes of K_6. The calculation is finite. An application of (3.4) deletes one edge and contributes a factor p, while an application of (3.5) deletes the endpoints of one edge and contributes a factor of order pN. We may choose the second operation at any edge, whereas the first operation requires a bound after each possible choice of the last edge.

The bound we seek for a typical nine-edge graph H is

$$t(H, p, N) = O(p^4 N^3).$$

Indeed, the six unqueried edges of a completed K_6 then contribute another factor p^6, giving $p^{10} N^3 = \lambda^3$. To see exactly which graphs can be handled in this way, we record the best monomial obtained by the recursions. For the moment we ignore constant factors and the binomial tail in (3.5); both will be restored at the end.

For an integer $b \ge 0$, let $A_b(H)$ be the largest integer a for which these recurrences certify

$$t(H, p, N) = O_H(p^a N^b).$$

The empty-graph base case is

$$A_b(V, \varnothing) = \begin{cases} 0, & b \ge |V|, \\ -\infty, & b < |V|. \end{cases}$$

For $E(H) \ne \varnothing$, the two recurrences give

$$A_b(H) = \max \left\{ 1 + \min_{e \in E(H)} A_b(H - e),\ 1 + \max_{uv \in E(H)} A_{b-1}(H - \{u, v\}) \right\}, \tag{5.1}$$

where the second term is omitted when $b = 0$, and $1 + (-\infty) = -\infty$. The first term takes the smallest exponent among the possible last edges, because all of them must be summed. The second term takes the best adjacent pair, because we are free to choose the pair in (3.5). Thus (5.1) records exactly the monomial bounds obtainable from the two inequalities. The fixed coefficient 2κ does not affect the exponent.

Lemma 4.1 (classification of the nine-edge graphs)**.** *Among the twenty-one isomorphism classes of nine-edge graphs on six vertices, the recurrence* (5.1) *with* $b = 3$ *gives the following distribution.*

$A_3(H)$	*Number of unlabeled types*	*Number of labeled graphs*
$-\infty$	2	75
3	1	360
4	10	3180
5	7	1380
6	1	10

The three types with $A_3(H) < 4$ are exactly

$$H_3, \qquad (K_5 - e) \sqcup K_1, \qquad B_4 = K_2 \vee \overline{K_4}.$$

Their degree sequences and labeled orbit sizes are

Graph	*Degree sequence*	*Orbit size*
H_3	$(5, 4, 3, 3, 2, 1)$	360
$(K_5 - e) \sqcup K_1$	$(4, 4, 4, 3, 3, 0)$	60
B_4	$(5, 5, 2, 2, 2, 2)$	15

For each of the remaining eighteen types, the recursive inequalities give

$$t(H, p, N) \leq C_H p^4 N^3 + R_H(p, N), \tag{5.2}$$

where, at the scale $N = \lfloor \lambda p^{-10/3} \rfloor$, $p^6 R_H(p, N) = o(1)$.

Proof. There are $\binom{15}{9} = 5005$ labeled graphs. We evaluate (5.1) for each of them using exact integer arithmetic and then identify graphs that differ only by a permutation of the six vertices. Appendix A gives the full reproducible calculation. The three isomorphism classes with $A_3(H) < 4$ are exactly the graphs displayed in the statement. Among the other eighteen classes, ten give $O(p^4N^3)$, seven give $O(p^5N^3)$, and one gives $O(p^6N^3)$.

Finally, every use of (3.5) also contributes a copy of the binomial tail τ_8. There are only finitely many recursion steps, so the total error R_H is at most a fixed polynomial in N and p^{-1} times $\exp(-cpN)$. Since $pN = \Theta(p^{-7/3})$, this gives $p^6 R_H = o(1)$, as required. $\square$

The following example illustrates the calculation. Let P_4 and C_4 be the path and cycle on four vertices. Successive applications of the two recursive inequalities give, up to exponentially small tails,

$$t(K_2) = O(pN), \qquad t(P_4) = O(p^2N^2), \qquad t(C_4) = O(p^3N^2).$$

Deleting the endpoints of any edge of $K_{3,3}$ leaves a C_4, and therefore

$$t(K_{3,3}) = O(pN)\, t(C_4) = O(p^4N^3).$$

The finite calculation repeats this argument, choosing the best sequence of recursive steps for each of the 5005 nine-edge graphs.

5 The lower bound

We now have all the ingredients needed for the lower bound. The argument is the same for each ordering of the edges of K_6, except that the three exceptional nine-edge graphs must be treated separately.

Proof of the lower bound in Theorem 1.1. Let $N = \lfloor \lambda p^{-10/3} \rfloor$, with $0 < \lambda \le 1$ fixed, and consider an arbitrary N-turn strategy on $[2N]$. Partition the completed labeled copies of K_6 according to their relative query order π.

Fix one order π. If $J_9(\pi)$ is one of the eighteen ordinary types in Lemma 4.1, then Lemmas 2.3 and 4.1 give

$$\mathbb{E}X_\pi \le p^6 t(J_9(\pi), p, N) \le Cp^{10}N^3 + o(1) \le C\lambda^3 + o(1). \tag{6.1}$$

Suppose next that $J_9(\pi)$ is exceptional. If $J_9(\pi) \cong H_3$, Corollary 3.4 gives

$$\mathbb{E}X_\pi \le p^6 t(H_3, p, N) \le C\lambda^3 + o(1). \tag{6.2}$$

If $J_9(\pi) \cong (K_5 - e) \sqcup K_1$, the uniform triple-codegree cutoff in Proposition 3.1 retains one factor λ^{-1}. Consequently,

$$\mathbb{E}X_\pi \le p^6 t((K_5 - e) \sqcup K_1, p, N) \le C\lambda^{-1}p^{10}N^3 + o(1) \le C\lambda^2 + o(1). \tag{6.3}$$

This is the only quadratic-in-λ main term.

The last case is

$$J_9(\pi) \cong B_4 = K_2 \vee \overline{K_4}.$$

The six edges missing from B_4 are exactly the six pairs among its four independent-side vertices. Therefore the tenth edge in the order necessarily produces

$$J_{10}(\pi) \cong Q = K_2 \vee (K_2 \sqcup 2K_1).$$

Stopping at ten edges rather than nine and applying Lemma 2.3 and Corollary 3.4,

$$\mathbb{E}X_\pi \le p^5 t(Q, p, N) \le C\lambda^3 + o(1). \tag{6.4}$$

There are only 15! orders. Summing (6.1)–(6.4) gives absolute constants C_2, C_3 such that

$$\mathbb{E}\, N_{K_6}(F_N) \le C_2\lambda^2 + C_3\lambda^3 + o(1). \tag{6.5}$$

Choose $\lambda > 0$ so small that $C_2\lambda^2 + C_3\lambda^3 < 1/2$. For all sufficiently small p, Markov's inequality then gives

$$\mathbb{P}(F_N \supseteq K_6) < \frac{1}{2}.$$

Proposition 2.1 transfers this conclusion to the original infinite-board game, proving $f(K_6, p) = \Omega(p^{-10/3})$. □

6 The branch and fill strategy

For completeness, we give the upper-bound construction of [3] in the special case needed here. In the notation of [3, Lemma 21], this is the choice $a = 1$, $b = 4$, and $\alpha = 2/3$. The strategy has the same three phases as in that paper: choose a root, build a large neighborhood of the root, and then repeatedly branch from that neighborhood and fill the resulting smaller set.

Let L and c_0 be sufficiently large constants, and put

$$T = \lceil Lp^{-10/3} \rceil, \qquad R = \lceil c_0 p^{-2/3} \rceil, \qquad s = \left\lfloor \frac{1}{8} p^2 T \right\rfloor.$$

Builder proceeds as follows.

1. Choose a vertex u, which will be the root of the final clique.
2. Query the pairs from u to T new vertices, and let W be the set of successful neighbors. If

$$\frac{1}{2}pT \le |W| \le 2pT, \tag{7.1}$$

keep at most $\lfloor 2pT \rfloor$ vertices of W; otherwise declare the attempt unsuccessful.
3. Starting with $U_1 = W$, perform R trials. In trial i, choose $w_i \in U_i$ and query all pairs from w_i to $U_i \setminus \{w_i\}$. Let Z_i be the number of successful answers. If $Z_i \ge s$, choose a set S_i of exactly s successful neighbors and query every pair inside S_i. We call the first set of queries the *branch* and the second the *fill*. After the trial, remove w_i and all its Z_i successful neighbors from the reservoir.

The reason for the construction is simple. If the graph built inside some S_i contains a K_4, then this K_4, together with u and w_i, forms a K_6. Every vertex of S_i is adjacent to u because it lies in W, and it is adjacent to w_i by the definition of the branch.

We next check that the strategy can carry out all R trials with high probability. Provided $|U_i| \ge pT/4$, conditional on all previous answers,

$$Z_i \sim \mathrm{Bin}(|U_i| - 1, p),$$

and the standard Chernoff bounds give, for a suitable absolute C_0,

$$\mathbb{P}(Z_i < s \text{ or } Z_i > C_0 p^2 T \mid \text{the past}) \le \exp(-cp^2T). \tag{7.2}$$

On the event that all previous trials satisfy this estimate, the number of vertices removed from W is at most

$$R(1 + C_0 p^2 T) = o(pT).$$

Thus (7.1) guarantees, by induction, that $|U_i| \ge pT/4$ throughout all R trials. Since $p^2T = \Theta(Lp^{-4/3})$, a union bound in (7.2) shows that the reservoir or branch conditions fail with probability $o(1)$. To give the strategy a deterministic query bound on every answer sequence, Builder simply aborts if one of these conditions fails or if $Z_i > C_0p^2T$. An aborted play may be padded with arbitrary new queries if one insists on exactly, rather than at most, the stated number of turns.

The number of queries is now easily bounded. The initial star costs T turns. Since $|U_i| \le 2pT$, all branches together cost at most

$$2RpT = O(p^{1/3}T),$$

and all fills together cost

$$R\binom{s}{2} = O(Rp^4T^2) = O(p^{-10/3}) = O(T).$$

Thus Builder uses $O(T)$ turns on every answer sequence.

It remains to show that one of the fills is likely to succeed. Conditional on the past and on a successful branch, the pairs inside S_i have not been queried before and form a random graph $G(s,p)$. Let X_i be its number of copies of K_4. Then

$$\mu := \mathbb{E}X_i = \Theta(p^6 s^4) = \Theta(L^4 p^{2/3}).$$

In the second moment, two distinct copies of K_4 can have dependent edge sets only if they share an edge. Copies sharing exactly an edge or a triangle contribute at most Cs^6p^{11} and Cs^5p^9, respectively. Hence

$$\mathbb{E}X_i^2 \le \mu + \mu^2 + C\big(s^6p^{11} + s^5p^9\big) \le C'\mu.$$

Here $s = \Theta(Lp^{-4/3})$, and the last inequality holds for all sufficiently small p. By the Paley–Zygmund inequality,

$$\mathbb{P}(X_i > 0 \mid \text{the past and a successful branch}) \ge c\mu = \Omega(L^4p^{2/3}).$$

The sets used in different trials are disjoint, so iterated conditioning shows that the probability that every fill fails is at most

$$(1 - cL^4p^{2/3})^R \le \exp(-cc_0L^4).$$

Taking L and c_0 sufficiently large makes this quantity smaller than $1/3$. The reservoir and branch failures contribute only $o(1)$, so the overall success probability is greater than $1/2$ for all sufficiently small p. Therefore

$$f(K_6, p) = O(p^{-10/3}).$$

Together with Section 5, this completes the proof of Theorem 1.1.

7 Why the tenth edge matters

It may be helpful to explain why the book B_4 is treated differently from the other nine-edge graphs. One might hope to stop every ordered copy of K_6 after its ninth edge and bound all nine-edge prefixes in the same way. This is false for B_4. Roughly speaking, Builder may construct about p^{-1} root edges and keep about p^2N common neighbors for each. This uses order pN successful edges and suggests order p^7N^4 rooted copies of B_4. When $N = p^{-10/3}$, this is larger than the desired bound p^4N^3 by a factor $p^{-1/3}$. This calculation is only a heuristic, but it identifies the obstruction correctly.

The obstruction disappears as soon as Builder makes progress toward a completed K_6. The next edge must join two of the four pages, so a successful answer creates Q. Moreover,

$$\sum_{uv \in E(F)} e(F[C_F(uv)]) = \Theta(N_{K_4}(F))$$

up to a fixed factor coming from labels. Thus copies of Q can be counted through copies of K_4, and the latter are only linear in the number of successfully built edges. This is why the proof stops after ten edges in the book case and after nine edges in every other case.

A The finite calculation

For completeness, we describe the exact calculation used in Lemma 4.1. Represent a graph on six labeled vertices by a bitmask on the fifteen possible edges, and evaluate the integer recurrence (5.1) by memoization. We do this for each of the 5005 masks having nine edges. The three exceptional isomorphism classes are then obtained by applying all 6! vertex permutations to explicit representatives. The resulting labeled histogram is

$$\begin{aligned}\#\{A_3=-\infty\} &= 75,\\ \#\{A_3=3\} &= 360,\\ \#\{A_3=4\} &= 3180,\\ \#\{A_3=5\} &= 1380,\\ \#\{A_3=6\} &= 10,\end{aligned}$$

and

$$\{H : A_3(H) < 4\} = \mathrm{Orb}(H_3) \cup \mathrm{Orb}((K_5 - e) \sqcup K_1) \cup \mathrm{Orb}(B_4).$$

The three orbit sizes $360 + 60 + 15 = 435$ agree with the two sub-threshold histogram entries. Quotienting by canonical relabeling gives the unlabeled histogram

$$2,\ 1,\ 10,\ 7,\ 1,$$

for a total of twenty-one types.

The accompanying exact-arithmetic program prints both histograms and the three orbit sizes, so the enumeration can be reproduced independently of the rest of the proof.

B Lean formalization and reproducibility

The proof has been formalized in Lean 4.24.0 [6], using Mathlib [17]. The complete source is available in the public repository [15]. This appendix records the precise scope of the formal result; none of the formalization is needed to read the combinatorial proof above.

B.1 Formal statement and conventions

For $p \in [0,1]$ and $N \in \mathbb{N}$, let $\mathsf{Win}_\infty(p,N)$ denote the existence of a deterministic adaptive strategy on the unordered pairs of $\mathbb{N}$ which makes exactly N distinct nonloop queries and finds a labeled copy of K_6 with probability at least $1/2$. Its probability is the finite Bernoulli sum over the 2^N possible answer strings. Thus the formalization uses precisely the finite-dimensional law seen by an N-query strategy; it does not require a construction of the infinite product probability space.

Likewise, let $\mathsf{Win}_{2N}(p,N)$ denote the corresponding statement for strategies whose vertex set is $[2N]$. The development proves, for every p and N,

$$\mathsf{Win}_\infty(p,N) \quad\Longleftrightarrow\quad \mathsf{Win}_{2N}(p,N). \tag{B.1}$$

In the source this is the theorem

`OnlineRamsey.InfinitePolicyBridge.infiniteAchievable_iff_achievable`.

The nontrivial direction of (B.1) formalizes the first-appearance argument from Proposition 2.1. At every node of the policy tree, vertices are named by the order in which they first occur in the transcript. Names assigned before a given history are unchanged when that history is extended. Histories with a common prefix therefore induce the same relabeling on that prefix, so the construction defines one coherent policy on $[2N]$, rather than a separate relabeling for each terminal transcript. The formal proof establishes the replay identity along every answer string, verifies freshness and the absence of loops on every history, and transports any labeled K_6 in the original transcript to one in the relabeled transcript. In the reverse direction, a policy on $[2N]$ is transported through the natural inclusion $[2N] \hookrightarrow \mathbb{N}$. The two constructions give the success-probability inequalities needed for (B.1), including the case $N = 0$.

Let

$$m_\infty(q) = \min\{N : \mathsf{Win}_\infty(q^3, N)\}.$$

The final checked theorem states that there is a constant $c > 0$ such that

$$c \le m_\infty(q)q^{10} \le 145160667136 \qquad (0 < q \le 1). \tag{B.2}$$

Its name in the source is

`OnlineRamsey.InfiniteUnconditional.exists_infiniteCubicPowerLaw`.

Equation (B.2) is a denominator-free version of the $p^{-10/3}$ power law: setting $p = q^3$ gives $q^{-10} = p^{-10/3}$. The use of exactly N queries agrees with the usual at-most-N convention by padding a shorter strategy with fresh queries. Randomized strategies give no smaller threshold, since one may condition on the private random seed and retain a deterministic strategy whose success probability is at least the average.

B.2 Scope and quantitative specialization

The checked dependency chain includes the finite classification in Appendix A, the adaptive Bernoulli product identities, the stopping-history argument, the random-host estimates, the ordinary and exceptional prefix cases, the lower-bound limiting argument, and a completely specified star–reservoir–branch–fill strategy for the upper bound. It also includes minimization and arbitrary-density rounding. The upper proof uses an explicit block schedule and conservative integer rounding, producing the constant

$$(16384 + 769 \cdot 184320)2^{10} = 145160667136;$$

no attempt was made to optimize it.

For comparison with the asymptotic notation in Section 3, the formal host argument uses $\ell = \lambda$, $p = q^3$, and the integer cutoffs

$$\begin{aligned} M &= \lceil 8q^3N \rceil, \qquad D = \lceil 4\sqrt{q^3M} \rceil, \qquad L = 90, \\ c_2 &= \lceil 12q^6N \rceil, \qquad c_3 = \left\lceil (560/\ell)q^9N \right\rceil. \end{aligned}$$

For each fixed $0 < \ell \le 1$, once q is sufficiently small, the side conditions

$$q \le \tfrac{1}{2}, \qquad 2q^{10} \le \ell, \qquad 40q^3 \le \ell, \qquad 160q \le 4\sqrt{8\ell/2}$$

hold, and the host-failure term is bounded by q^{70}. Since $N \le \ell q^{-10}$, the crude six-vertex bound gives

$$(2N)^6 q^{70} \le 64\ell^6 q^{10} \longrightarrow 0.$$

This is one explicit quantitative instantiation of Proposition 3.1.

B.3 Reproduction and trust

The source is available at https://github.com/wpoverman/k6-subgraph-query-lean. With Git and `elan` installed, a reader may obtain and check the development as follows.

```
git clone https://github.com/wpoverman/k6-subgraph-query-lean.git
cd k6-subgraph-query-lean
lake build
```

The project pins Lean to version 4.24.0 through `lean-toolchain`, and the Mathlib revision is recorded in `lake-manifest.json`; hence the build uses the versions specified by the repository rather than the user's global Lean default. After a successful build, the principal additional verification commands are

```
lake exe verify-k6-prefix
lake -R --no-cache build OnlineRamsey.InfiniteUnconditional
lake env lean TrustAudit.lean
```

The first of these commands prints the histograms and orbit sizes from Appendix A. The second rebuilds every dependency of the final theorem without using cached project objects, while the third displays the axioms on which the final theorem depends. For an exact record of the source revision used in a run, one may also execute

```
git rev-parse HEAD
```

inside the project directory. A tagged archival release, preferably with a DOI, would provide an immutable reference for future versions of the paper.

The checked source contains no `sorry`, `admit`, or user-declared axioms. The trust audit for the final theorem reports the standard principles `propext`, `Classical.choice`, and `Quot.sound`. The exhaustive finite classification is discharged by `native_decide`, which additionally introduces `Lean.ofReduceBool` and `Lean.trustCompiler`, as usual for native-decision certificates.

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